

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 0b5be785 · Data hash 349e...737e

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 07:14 and 07:27 UTC. Reported results: test_loss 5, test_ppl 148.78. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	0b5be785-ba14-4542-9378-46738e8a1d1d
Initialized	2026-10-07T07:14:13Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	6bbc374f8cea748b956d6d5b3893a33ef70792d281ee8d4362bf7f6b36a5b14c
Disclosure	open
Source code, run 1	faf55397193f6022eea300e30204d49aa2ecf36e676151e158d4596e72d29b52

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	13 minutes 6 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
valid_loss	5.07	20 (20 readings)	explicit
valid_ppl	159.49	20 (20 readings)	explicit

Inline claim	Value	Phase	Output line
test_loss	5		391
test_ppl	148.78		392

Seed committed: seed:1111, output line 1, 2026-10-07T07:14:14Z.

4 Execution evidence

Coherence

Coherent (0.37): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	796.0 s	1.47 GB	91%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 121 tracked files, 4 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	07:14:14	a253305778d4ae20	121
3	run_end	07:27:20	a253305778d4ae20	0

2 snapshots with no file change omitted here; full timeline in Appendix B.

Activity: 124 files read, 0 files written, 2 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T07:27:22.297167+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	349ed2af849a2c67dbe68b1d153cedcb4523a1c5ebdd45cc3f17dd7d64e7737e
Code bundle hash	a213f90f61722cf9b42ceda507608b90af82eb65cf5332f439b53f385eae1b2a

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
valid_loss	5.75	1	24	explicit
valid_ppl	313.69	1	25	explicit
valid_loss	5.51	2	43	explicit
valid_ppl	247.16	2	44	explicit
valid_loss	5.4	3	62	explicit
valid_ppl	222.2	3	63	explicit
valid_loss	5.36	4	81	explicit
valid_ppl	212.61	4	82	explicit
valid_loss	5.34	5	100	explicit
valid_ppl	208.65	5	101	explicit
valid_loss	5.3	6	119	explicit
valid_ppl	200.03	6	120	explicit
valid_loss	5.31	7	138	explicit
valid_ppl	201.66	7	139	explicit
valid_loss	5.2	8	157	explicit
valid_ppl	180.87	8	158	explicit
valid_loss	5.2	9	176	explicit
valid_ppl	180.93	9	177	explicit
valid_loss	5.14	10	195	explicit
valid_ppl	171.05	10	196	explicit
valid_loss	5.14	11	214	explicit
valid_ppl	171.06	11	215	explicit
valid_loss	5.11	12	233	explicit
valid_ppl	164.87	12	234	explicit
valid_loss	5.1	13	252	explicit
valid_ppl	164.59	13	253	explicit
valid_loss	5.1	14	271	explicit
valid_ppl	164.58	14	272	explicit
valid_loss	5.1	15	290	explicit
valid_ppl	164.68	15	291	explicit
valid_loss	5.07	16	309	explicit
valid_ppl	159.87	16	310	explicit
valid_loss	5.07	17	328	explicit
valid_ppl	159.69	17	329	explicit
valid_loss	5.07	18	347	explicit
valid_ppl	159.62	18	348	explicit
valid_loss	5.07	19	366	explicit
valid_ppl	159.51	19	367	explicit
valid_loss	5.07	20	385	explicit
valid_ppl	159.49	20	386	explicit

Phase	Line	Time	CPU time	Memory	GPU
train	2	07:14:14	0.4 s	0.54 GB	
test	390	07:27:20	795.6 s	1.31 GB	60%, 566 MB

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	07:14:14	a253305778d4ae20	121
1	phase train	07:14:14	a253305778d4ae20	0
2	phase test	07:27:20	a253305778d4ae20	0
3	run_end	07:27:20	a253305778d4ae20	0

Event	Path or command	Hash
read	kv_run.sh	
read	word_language_model/main.py	
read	.github/workflows/doc-build.yml	
read	.github/workflows/main_cpp.yml	
read	.github/workflows/main_distributed.yml	
read	.github/workflows/main_python.yml	
read	cpp/autograd/autograd.cpp	
read	cpp/custom-dataset/custom-dataset.cpp	
read	cpp/dcgan/dcgan.cpp	
read	cpp/dcgan/display_samples.py	
read	cpp/distributed/dist-mnist.cpp	
read	cpp/mnist/mnist.cpp	
read	cpp/regression/regression.cpp	
read	cpp/tools/download_mnist.py	
read	cpp/transfer-learning/classify.cpp	
read	cpp/transfer-learning/convert.py	
read	cpp/transfer-learning/main.cpp	
read	cpp/transfer-learning/main.h	
read	dcgan/main.py	
read	distributed/FSDP/T5_training.py	
read	distributed/FSDP/configs/__init__.py	
read	distributed/FSDP/configs/fsdp.py	
read	distributed/FSDP/configs/training.py	
read	distributed/FSDP/download_dataset.sh	
read	distributed/FSDP/model_checkpointing/__init__.py	
read	distributed/FSDP/model_checkpointing/checkpoint_handler.py	
read	distributed/FSDP/policies/__init__.py	
read	distributed/FSDP/policies/activation_checkpointing_functions.py	
read	distributed/FSDP/policies/mixed_precision.py	
read	distributed/FSDP/policies/wrapping.py	
read	distributed/FSDP/summarization_dataset.py	
read	distributed/FSDP/utils/__init__.py	
read	distributed/FSDP/utils/environment.py	
read	distributed/FSDP/utils/train_utils.py	
read	distributed/FSDP2/checkpoint.py	
read	distributed/FSDP2/example.py	
read	distributed/FSDP2/model.py	
read	distributed/FSDP2/run_example.sh	
read	distributed/FSDP2/utils.py	
read	distributed/ddp-tutorial-series/datautils.py	

Event	Path or command	Hash
read	distributed/ddp-tutorial-series/multigpu.py	
read	distributed/ddp-tutorial-series/multigpu_torchrun.py	
read	distributed/ddp-tutorial-series/multinode.py	
read	distributed/ddp-tutorial-series/single_gpu.py	
read	distributed/ddp-tutorial-series/slurm/sbatch_run.sh	
read	distributed/ddp/example.py	
read	distributed/ddp/run_example.sh	
read	distributed/minGPT-ddp/mingpt/char_dataset.py	
read	distributed/minGPT-ddp/mingpt/gpt2_train_cfg.yaml	
read	distributed/minGPT-ddp/mingpt/main.py	
read	distributed/minGPT-ddp/mingpt/model.py	
read	distributed/minGPT-ddp/mingpt/slurm/sbatch_run.sh	
read	distributed/minGPT-ddp/mingpt/trainer.py	
read	distributed/minGPT-ddp/run_example.sh	
read	distributed/rpc/batch/parameter_server.py	
read	distributed/rpc/batch/reinforce.py	
read	distributed/rpc/ddp_rpc/main.py	
read	distributed/rpc/parameter_server/rpc_parameter_server.py	
read	distributed/rpc/pipeline/main.py	
read	distributed/rpc/rl/main.py	
read	distributed/rpc/rnn/main.py	
read	distributed/rpc/rnn/rnn.py	
read	distributed/tensor_parallelism/fsdp_tp_example.py	
read	distributed/tensor_parallelism/llama2_model.py	
read	distributed/tensor_parallelism/log_utils.py	
read	distributed/tensor_parallelism/run_example.sh	
read	distributed/tensor_parallelism/sequence_parallel_example.py	
read	distributed/tensor_parallelism/tensor_parallel_example.py	
read	docs/source/conf.py	
read	fast_neural_style/download_saved_models.py	
read	fast_neural_style/neural_style/__init__.py	
read	fast_neural_style/neural_style/neural_style.py	
read	fast_neural_style/neural_style/transformer_net.py	
read	fast_neural_style/neural_style/utils.py	
read	fast_neural_style/neural_style/vgg.py	
read	fx/custom_tracer.py	
read	fx/inline_function.py	
read	fx/invert.py	
read	fx/module_tracer.py	
read	fx/native_interpreter/interpreter.cpp	
read	fx/native_interpreter/use_interpreter.py	
read	fx/primitive_library.py	
read	fx/profiling_tracer.py	
read	fx/proxy_based_graph_creation.py	
read	fx/replace_op.py	
read	fx/subgraph_rewriter_basic_use.py	
read	fx/wrap_output_dynamically.py	
read	gat/main.py	
read	gcn/main.py	
read	imagenet/extract_ILSVRC.sh	

Event	Path or command	Hash
read	imagenet/main.py	
read	language_translation/main.py	
read	language_translation/src/data.py	
read	language_translation/src/model.py	
read	legacy/snli/model.py	
read	legacy/snli/train.py	
read	legacy/snli/util.py	
read	mnist/main.py	
read	mnist_forward_forward/main.py	
read	mnist_hogwild/main.py	
read	mnist_hogwild/train.py	
read	mnist_rnn/main.py	
read	regression/main.py	
read	reinforcement_learning/actor_critic.py	
read	reinforcement_learning/reinforce.py	
read	run_cpp_examples.sh	
read	run_distributed_examples.sh	
read	run_python_examples.sh	
read	siamese_network/main.py	
read	super_resolution/data.py	
read	super_resolution/dataset.py	
read	super_resolution/main.py	
read	super_resolution/model.py	
read	super_resolution/super_resolve.py	
read	time_sequence_prediction/generate_sine_wave.py	
read	time_sequence_prediction/train.py	
read	utils.sh	
read	vae/main.py	
read	word_language_model/generate.py	
read	word_language_model/model.py	
read	word_language_model/data.py	
read	word_language_model/data/wikitext-2/train.txt	
read	word_language_model/data/wikitext-2/valid.txt	
read	word_language_model/data/wikitext-2/test.txt	
process	../env/bin/python -u main.py --accel --epochs 20 --model Transformer --lr 5 ...	
process	awk { print; fflush() } /^\\ end of epoch/ { match(\$0, /epoch +[0-9]+/); spli...	

C Run history

Invocation 1, OK, 13 minutes 6 seconds, 393 output lines, started 2026-10-07T07:14:14

metric_hash 963a01f365b08b3c69aa0cc349ec9072e7d56f2baa2220efdd12f9f97fd9b0b6
run_digest bee29d98999beb20e86599faa2c48a6eea846beaacea10706aa58fb3b68b5c91
anchored at run end

D Raw seal fields and hashes

Seal

```
data_hash      349ed2af849a2c67dbe68b1d153cedcb4523a1c5ebdd45cc3f17dd7d64e7737e
signed_message_hash 3de55a112d35c0be5b7a9886d7e12f6ffcc27bd5e8a1adb3c130822894348499
nonce         b62713749a6ce5c54073244e530def23
signed_at     2026-10-07T07:27:22.297167+00:00
sealed_at     2026-10-07T07:27:22.355769266Z
checkout_bundle  a213f90f61722cf9b42ceda507608b90af82eb65cf5332f439b53f385eae1b2a
```

Signature (base64)

```
XUVUg1vs5GszvPA3DLVK46e7cIbaylII/FdJM13ZKKJTD7k+SSfXQM9QPlwB6sjj1y8vjnuhwyuC6kn8NkYvghIj
AxTc5RucHy1Va8m1+j0luOgzt+ttNVsbP5jjPv8gllf6A9Wi5qZTYEuPwGoJiUI4II0bRsuo9124XE02aL0N52tw
tDZqi5MPIMqjll/44Q2zFS93pCofW5MUIooLi8+SXvYTSeTh1IyDZWNCxq+WZ43pCQ6oMBQ3y2sw+4T8LcTv+L1W
JBYvz/Tpk2KNRvUpDaVto0Tu/XGRwN8R1KbKCEpCr31ZyenEf1DgA6dqJ+Fof1K2rKPk4frAEMvF2NTdvY6fFFxf
KBmKMx4Qck53F00gDvbEycKHHUNGDFLuRwlb/zJoYw6mLJp/oSL4X4pw+GdBBioMfEskEv8/jCZisHAW99n9MnnE
nBtLLO5EjPFDbKR5oG7/BRugWlk8IioqqfaYxa0vsJSTZB36sz2Ze2cd8eYsiLN/iS0xR5wtHpsvLUHgoEpN+zja
yv8IY2ytoVI5EF+prxt2TWA0nBQF9E984GPK4IoQrv3ctNBEC4nubpkhCy0/76U/h3zrm/1m248mwUrr6x0G1ZTj
gwuvjHBmNY+PvfQn0zATNEGWHCaL/2MogKRJEoRukHYLCb2pECu9myMLbFVj8WohCs=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICIjANBgqhkiG9w0BAQEFAAOCAg8AMIICCGKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlP1cjayJxD5fcProPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAT4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNpxYa5X
7qU95EFEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeadbIbHN
av0JjdTJ//CMOWZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdhQ+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9IL5xEaJn4bku4eKqeS7VSzt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPGpZ242NH8JgE6CNF7ToL9syans67u7UMgIcNgeNHnCy9bLgPU5annzP9gN2l
6HEMLdkXNdKQRb1rDWYEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7LE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCKDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdW3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout  efcc0f1645964ca7b7102d20fda1abb2b3cfaaf37a4de68f7e9edd16a3d6adef
stderr  e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 963a01f365b08b3c69aa0cc349ec9072e7d56f2baa2220efdd12f9f97fd9b0b6
digest  bee29d98999beb20e86599faa2c48a6eea846beaacea10706aa58fb3b68b5c91
kv_run.sh (unchanged)
pre 9369a17f31da0ffa7d9ea70792027fed9ae68b855e8888822038b18e036e4ac4
post 9369a17f31da0ffa7d9ea70792027fed9ae68b855e8888822038b18e036e4ac4
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```